



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS ADVANCED

2021 Year 12 Course Assessment Task 4 (Trial HSC Examination)

Monday, 30 August 2021

General instructions

- Working time – 3 hour.
(plus 10 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- Attempt **all** questions.

SECTION I

- Mark your answers on the answer grid provided (on page 27)

SECTION II

- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

NESA STUDENT #: # BOOKLETS USED:

Class (please ✓)

☐ 12MAA.1 – Miss Lee

☐ 12MAX.3 – Mr Lam

☐ 12MAX.1 – Mr Ho

☐ 12MAX.2 – Mrs Bhamra

☐ 12MAX.4 – Mr Sun

Marker's use only.

QUESTION	1-10	11-15	16-18	19-21	22-25	26-29	30-32	Total	%
MARKS	$\overline{10}$	$\overline{15}$	$\overline{14}$	$\overline{15}$	$\overline{16}$	$\overline{17}$	$\overline{13}$	$\overline{100}$	

Section I

10 marks

Attempt Question 1 to 10

Allow approximately 15 minutes for this section

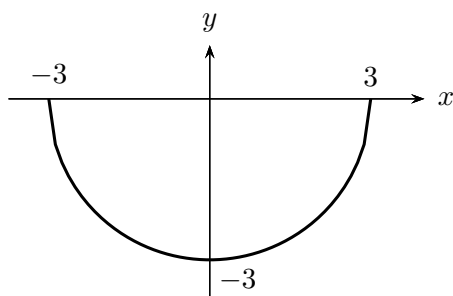
Mark your answers on the answer grid provided (labelled as page 27).

Questions

Marks

1. The equation of the semi-circle illustrated below is given by

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(A) $y = \sqrt{9 - x^2}$

(C) $y = \sqrt{3 - x^2}$

(B) $y = -\sqrt{9 - x^2}$

(D) $y = -\sqrt{3 - x^2}$

2. The period of the function $f(x) = \sin\left(3x - \frac{\pi}{3}\right)$, where $x \in \mathbb{R}$, is

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(A) $\frac{\pi}{9}$

(C) 2π

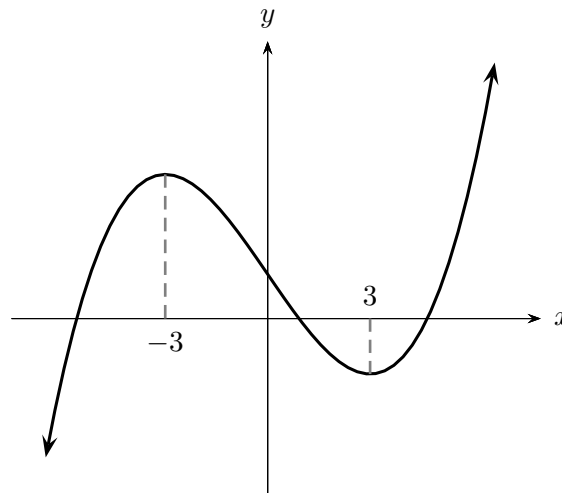
(B) $\frac{2\pi}{3}$

(D) $\frac{\pi}{3}$

Examination continues overleaf...

3. The graph of $y = f(x)$ is shown below. For which interval(s) is $f'(x) < 0$?

1



- (A) $(-\infty, -3) \cup (3, \infty)$ (C) $(-\infty, -3] \cup [3, \infty)$
(B) $(-3, 3)$ (D) $[-3, 3]$

4. Which of the following is $e^{2\log_e x}$ equal to?

1

- (A) $\sqrt{x}$ (C) $2x$
(B) x^2 (D) e^{2x}

5. A particle, initially at rest at the origin, moves in a straight line with velocity $v = 6 - 2t$ m/s.

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What is the total distance travelled by the particle in the first 6 seconds?

- (A) 0 m (C) 9 m
(B) 6 m (D) 18 m

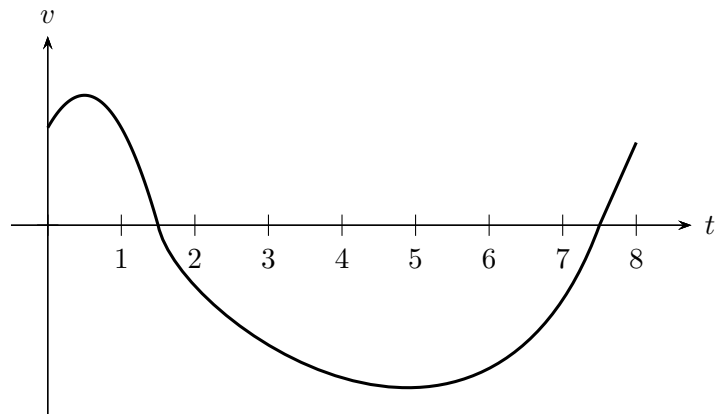
Examination continues overleaf...

6. If $\int_1^4 f(x) dx = 2$, which of the following is $\int_1^4 (2f(x) + 3) dx$ equal to? 1
- (A) 2 (C) 7
- (B) 13 (D) 10
7. In a class of 30 girls, 13 are dancers and 23 are gymnasts. If 7 girls do both dance and gymnastics, what is the probability that a girl chosen at random does neither dance nor gymnastics? 1
- (A) $\frac{1}{5}$ (C) $\frac{1}{30}$
- (B) $\frac{7}{30}$ (D) $\frac{8}{15}$
8. The circle $x^2 + y^2 - 4x - 4y + c = 0$ touches both the x and y axes. What is the value of c ? 1
- (A) $c = 1$ (C) $c = 4$
- (B) $c = 2$ (D) $c = 8$
9. The amount of water that Eleanor uses to wash her car is normally distributed with a mean of 50 litres and a standard deviation of 4 litres. 1
- On what percentage of occasions would Eleanor expect to use between 42 litres and 46 litres of water to wash her car?
- (A) 13.5% (C) 34%
- (B) 27% (D) 68%

Examination continues overleaf...

10. A particle is moving along the x -axis. The graph shows its velocity v metres per second at time t seconds.

1



Initially, the particle is at the origin.

Which of the following describes the position of the particle after 8 seconds?

- (A) The particle is to the left of the origin.
- (B) The particle is to the right of the origin.
- (C) The particle is at the origin.
- (D) There is not enough information to determine its position.

Examination continues overleaf...

Section II

90 marks

Attempt Question 11 to 32

Allow approximately 2 hours and 45 minutes for this section

Write your answers in the space provided.

Question 11 (2 Marks)

Consider the polynomial $P(x) = (x + 1)(x - 1)(2x - 5)$.

- (a) State the degree of $P(x)$. **1**

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- (b) State whether $P(x)$ is one-to-one, one-to-many, many-to-one or many-to-many. **1**

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Question 12 (3 Marks)

Differentiate the following:

- (a) $y = (3x^2 - x + 1)^5$ **2**

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- (b) $y = 2e^{x^2+x+1}$ **1**

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Examination continues overleaf...

Question 13 (4 Marks)

Find the following:

(a) $\int \left(x + \frac{1}{x}\right) dx$ **2**

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(b) $\int x e^{x^2+1} dx$ **2**

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Question 14 (3 Marks)

(a) Differentiate $\frac{x}{1 + \ln x}$. **2**

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(b) Hence, or otherwise, find the primitive of $\frac{\ln x}{(1 + \ln x)^2}$. **1**

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Examination continues overleaf...

Question 15 (3 Marks)

Find the equation of the normal to $y = x \sin x$ at the point where $x = \frac{\pi}{2}$.

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Examination continues overleaf...

Question 16 (5 Marks)

The number N of bacteria in a mouldy loaf of bread at time t hours is given by the equation

$$N = 21e^{kt}$$

After 7 hours, the number of bacteria present is 30.

- (a) Find the exact value of k .

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- (b) Determine the number of bacteria after 24 hours.

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- (c) At what rate is the number of bacteria increasing after 24 hours? Give your answer to 1 decimal place.

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Examination continues overleaf...

Question 17 (6 Marks)

The acceleration of a moving body is given by $a = \sqrt{2t + 1} \text{ ms}^{-2}$.

- (a) If the body starts from rest, find its velocity after 4 seconds. **3**

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- (b) Find, in exact form, the average velocity of the body during the fourth second. **3**

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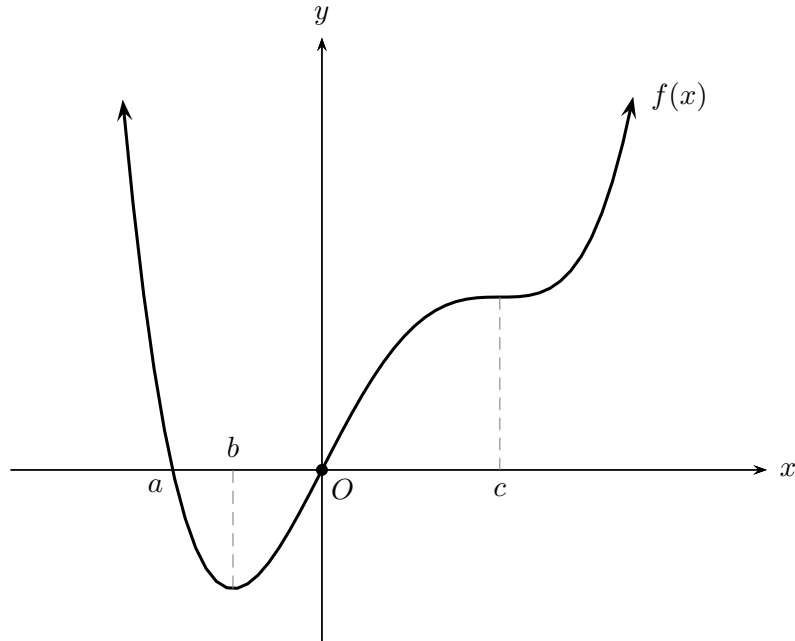
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Question 18 (3 Marks)

The graph of $f(x)$ is shown below. There is a minimum turning point at $x = b$, an inflexion point at O , and a horizontal point of inflexion at $x = c$.

3

Draw the graph of $f'(x)$ on the same set of axes below.



Examination continues overleaf...

Question 19 (5 Marks)

- (a) Use the trapezoidal rule with 5 function values to find an approximation to the value of $\int_1^5 \ln x \, dx$, correct to 4 decimal places. **3**

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- (b) With the aid of a sketch, explain whether the approximation found in part (a) is an over-approximation or an under-approximation. **2**

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Examination continues overleaf...

Question 20 (5 Marks)

A university student plans to finish university in two years time. He wishes to travel overseas and thinks he will need \$15000. He decides to invest \$ M at the start of each month in an investment account earning 3% per annum compounded monthly.

Let A_n be the amount at the end of n months.

- (a) Find A_2 in terms of M .

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- (b) Show that $A_n = 401M(1.0025^n - 1)$.

2

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- (c) Find the amount he must invest each month to reach his target of \$15000. Give your answer to the nearest dollar.

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Examination continues overleaf...

Question 21 (5 Marks)

Sketch the graph of the curve $y = x^4 + 2x^3 + 1$ for $-2 \leq x \leq 1$, labelling the stationary points and points of inflexion. Also state the global maximum in the specified domain.

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(Do NOT determine the x intercepts of the curve.)

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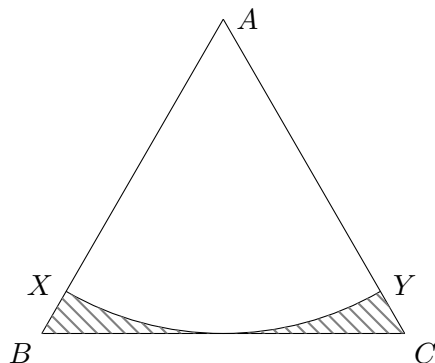
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Question 22 (6 Marks)

In the diagram, $\triangle ABC$ is an equilateral triangle with sides of length 6 cm. An arc with centre A and BC as tangent, cuts AB and AC at X and Y respectively.



- (a) Show that the radius of the arc is $3\sqrt{3}$ cm.

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- (b) Find the exact length of arc XY .

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- (c) Find in exact form, the area of the shaded region.

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Examination continues overleaf...

Question 23 (2 Marks)Find the value(s) of $\tan x$ when $\tan^2 x + \sec^2 x = 9$.**2**

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Question 24 (2 Marks)Prove that $\sec^2 \theta + \operatorname{cosec}^2 \theta = \sec^2 \theta \operatorname{cosec}^2 \theta$.**2**

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Examination continues overleaf...

Question 25 (6 Marks)

A game consists of a player tossing a fair coin three times. You lose \$3.00 if three heads appear and lose \$2.00 if two heads appear. You win \$1.00 if one head appears and win \$3.00 if no heads appear. Let X be the amount you win or lose per game.

- (a) Complete the probability distribution table below.

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x	-3	-2	1	3
$P(X = x)$				

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- (b) Calculate $E(X)$ and hence determine the expected profit or loss if you played the game 1000 times.

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- (c) Evaluate $P(X > -2 | X \geq -2)$.

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Examination continues overleaf...

Question 26 (4 Marks)

A continuous random variable T , represents the time taken in days to show symptoms after contracting a virus. It has the following probability density function

$$f(t) = \begin{cases} \frac{k}{2t-1} & \text{for } 1 \leq t \leq 14 \\ 0 & \text{for } 0 \leq t < 1 \text{ or } t > 14 \end{cases}$$

- (a) Show that $k = \frac{2}{\ln 27}$.

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Examination continues overleaf...

- (b) After how many days will a person have a 75% chance of showing symptoms after they have contracted the virus?

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Examination continues overleaf...

Question 27 (3 Marks)

Give a possible sequence of transformations that transforms $y = \frac{1}{x}$ to $y = \frac{3}{1-2x} + 5$. **3**
Ensure you use appropriate terminology in your response.

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Examination continues overleaf...

Question 28 (6 Marks)

- (a) Sketch the graph of $y = |2x - 3|$ on the lines provided below. Label the x and y intercepts. **2**

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- (b) Hence, or otherwise, solve $|6x - 9| + 5 = 8$. **3**

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- (c) By using the graph sketched in part (a), or otherwise, state the values of m such that $|2x - 3| = mx$ has exactly 1 real solution. **1**

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Examination continues overleaf...

Question 29 (4 Marks)

Suppose $f(x) = \frac{1}{x^2 + x - 2}$ and $g(x) = \sin x$.

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Find the vertical asymptotes of $y = f(g(x))$ within the domain $[0, 2\pi]$.

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Examination continues overleaf...

Question 30 (5 Marks)

There are 4 red and 3 black discs in a bag. Sophie and Emma are playing a game in which they take turns drawing a disc from the bag and then replacing it.

To win the game, Sophie must draw a red disc and for Emma to win she must draw a black disc. They continue taking turns until there is a winner. Sophie goes first.

- (a) Find the probability that Sophie wins on her first draw. **1**

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- (b) Find the probability that Sophie wins in three or less of her turns. **2**

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- (c) Find the probability that Sophie wins the game. **2**

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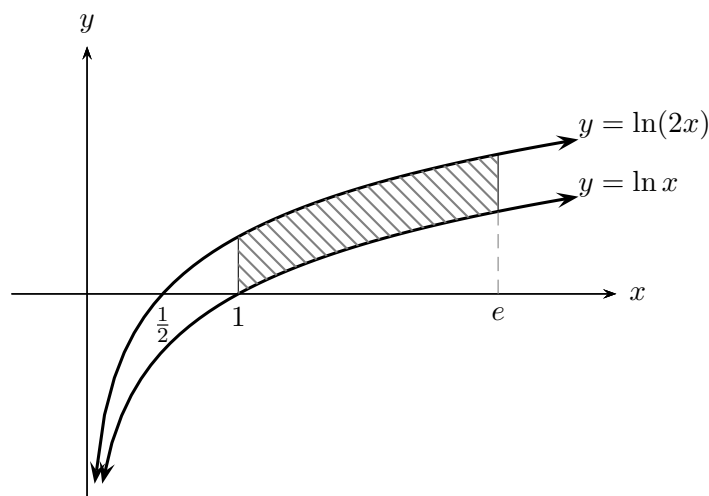
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Examination continues overleaf...

Question 31 (3 Marks)

The curves of $y = \ln x$ and $y = \ln(2x)$ are drawn below.

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Find the shaded area, in exact form, between the curves $y = \ln(2x)$ and $y = \ln x$ and the lines $x = 1$ and $x = e$.

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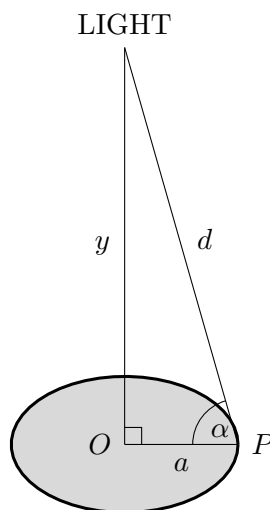
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Examination continues overleaf...

Question 32 (5 Marks)

A light is to be placed over the centre of a circle, radius a units.

The intensity, I , of the light is proportional to the sine of the angle, α , at which the rays strike the circumference of the circle, divided by the square of the distance, d , from the light to the circumference of the circle; i.e. $I = \frac{k \sin \alpha}{d^2}$, where k is a constant.



- (a) Show that $I = \frac{ky}{(y^2 + a^2)^{\frac{3}{2}}}$.

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Examination continues overleaf...

- (b) Find the best height for the light to be placed over the centre of the circle so as to provide the maximum illumination to the circumference. **3**

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End of paper.

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g. “●”

NESA STUDENT #:

Class (please ✓)

☐ 12MAA.1 – Miss Lee

☐ 12MAX.3 – Mr Lam

☐ 12MAX.1 – Mr Ho

☐ 12MAX.2 – Mrs Bhamra

☐ 12MAX.4 – Mr Sun

Directions for multiple choice answers

- Read each question and its suggested answers.
- Select the alternative (A), (B), (C), or (D) that best answers the question.
- Mark only one circle per question. There is only *one* correct choice per question.
- Fill in the response circle completely, using blue or black pen, e.g.

(A) (B) ● (D)

- If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

(A) (B) ~~●~~ ●

- If you continue to change your mind, write the word **correct** and clearly indicate your final choice with an arrow as shown below:

(A) (B) ~~●~~ ~~●~~ ^{correct}

1 – (A) (B) (C) (D)

2 – (A) (B) (C) (D)

3 – (A) (B) (C) (D)

4 – (A) (B) (C) (D)

5 – (A) (B) (C) (D)

6 – (A) (B) (C) (D)

7 – (A) (B) (C) (D)

8 – (A) (B) (C) (D)

9 – (A) (B) (C) (D)

10 – (A) (B) (C) (D)

Sample Band 6 Responses

Section I

1. (B) 2. (B) 3. (B) 4. (B) 5. (D)
6. (B) 7. (C) 8. (C) 9. (A) 10. (A)

Section II

Question 11 (Bhamra)

- (a) (1 mark)
✓ [1] for correct degree
Degree = 3.
- (b) (1 mark)
✓ [1] for correct classification
 $P(x)$ is many-to-one.

Question 12 (Bhamra)

- (a) (2 marks)
✓ [1] for obtaining $5(3x^2 - x + 1)^4$
✓ [1] for correct application of chain rule

$$\begin{aligned}\frac{dy}{dx} &= 5(3x^2 - x + 1)^4 \times (6x - 1) \\ &= 5(6x - 1)(3x^2 - x + 1)^4\end{aligned}$$

- (b) (1 mark)
✓ [1] for correct derivative

$$\frac{dy}{dx} = 2(2x + 1)e^{x^2+x+1}$$

Question 13 (Bhamra)

- (a) (2 marks)
✓ [1] for correct integration of x OR $\frac{1}{x}$
✓ [1] for correct primitive (including $+C$)

$$\int \left(x + \frac{1}{x}\right) dx = \frac{1}{2}x^2 + \ln x + C$$

- (b) (2 marks)
✓ [1] for obtaining an equivalent integral with $2x$
✓ [1] for correct primitive (including $+C$)

$$\begin{aligned}\int xe^{x^2+1} dx &= \frac{1}{2} \int 2xe^{x^2+1} dx \\ &= \frac{1}{2} e^{x^2+1} + C\end{aligned}$$

Question 14 (Bhamra)

- (a) (2 marks)
✓ [1] for correctly applying the quotient rule
✓ [1] for correct, fully simplified, derivative

$$\begin{aligned}\frac{d}{dx} \left(\frac{x}{1 + \ln x} \right) &= \frac{(1 + \ln x)(1) - x(\frac{1}{x})}{(1 + \ln x)^2} \\ &= \frac{\ln x}{(1 + \ln x)^2}\end{aligned}$$

- (b) (1 mark)
✓ [1] for correct primitive (including $+C$)

$$\int \frac{\ln x}{(1 + \ln x)^2} dx = \frac{x}{1 + \ln x} + C$$

Question 15 (Bhamra) (3 marks)

- ✓ [1] for correct derivative
✓ [1] for finding gradient of normal at $x = \frac{\pi}{2}$
✓ [1] for correct equation of normal

$$\frac{dy}{dx} = x \cos x + \sin x$$

$$\text{At } x = \frac{\pi}{2},$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{\pi}{2} \cos \frac{\pi}{2} + \sin \frac{\pi}{2} \\ &= 1\end{aligned}$$

$$\therefore m_N = -1$$

$\therefore$ Equation of normal at $\left(\frac{\pi}{2}, \frac{\pi}{2}\right)$:

$$\begin{aligned}y - \frac{\pi}{2} &= -\left(x - \frac{\pi}{2}\right) \\ \therefore y &= -x + \pi\end{aligned}$$

Question 16 (Ho)

(a) (2 marks)

✓ [1] for obtaining $30 = 21e^{7k}$ ✓ [1] for correct exact value of k When $t = 7$, $N = 30$

$$\begin{aligned}\therefore 30 &= 21e^{7k} \\ e^{7k} &= \frac{10}{7} \\ 7k &= \ln \frac{10}{7} \\ \therefore k &= \frac{1}{7} \ln \frac{10}{7}\end{aligned}$$

(b) (1 mark)

✓ [1] for correct value of N (rounding errors are ignored)When $t = 24$,

$$\begin{aligned}N &= 21e^{24k} \\ &= 71.336... \\ &\approx 71 \text{ bacteria}\end{aligned}$$

(c) (2 marks)

✓ [1] for correct expression of $\frac{dN}{dt}$ ✓ [1] for correct rate when $t = 24$ (rounding errors are ignored)

$$\frac{dN}{dt} = 21ke^{kt}$$

When $t = 24$,

$$\begin{aligned}\frac{dN}{dt} &= 21ke^{24k} \\ &= 3.634... \\ &\approx 3.6 \text{ bacteria/hr}\end{aligned}$$

Question 17 (Ho)

(a) (3 marks)

- ✓ [1] for obtaining $v = \frac{(2t+1)^{\frac{3}{2}}}{3} + C$
- ✓ [1] for finding the value of C
- ✓ [1] for finding v when $t = 4$

$$\begin{aligned}
 a &= (2t+1)^{\frac{1}{2}} \\
 \therefore v &= \int (2t+1)^{\frac{1}{2}} dt \\
 &= \frac{(2t+1)^{\frac{3}{2}}}{\frac{3}{2} \times 2} + C \\
 &= \frac{(2t+1)^{\frac{3}{2}}}{3} + C
 \end{aligned}$$

When $t = 0$, $v = 0$

$$\begin{aligned}
 \therefore 0 &= \frac{1}{3} + C \\
 \therefore C &= -\frac{1}{3} \\
 \therefore v &= \frac{(2t+1)^{\frac{3}{2}}}{3} - \frac{1}{3}
 \end{aligned}$$

When $t = 4$,

$$\begin{aligned}
 v &= \frac{9^{\frac{3}{2}}}{3} - \frac{1}{3} \\
 &= \frac{27}{3} - \frac{1}{3} \\
 &= \frac{26}{3} \text{ m/s}
 \end{aligned}$$

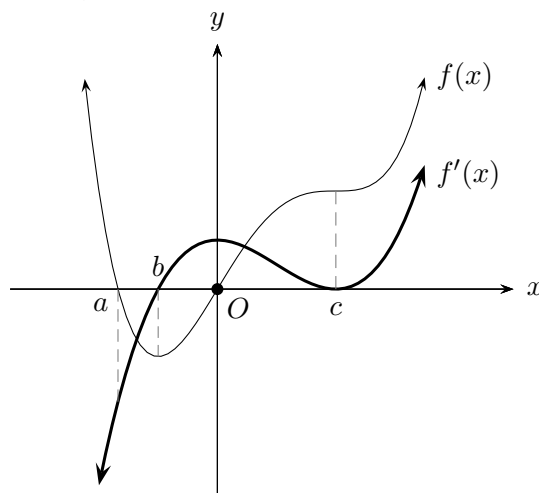
(b) (3 marks)

- ✓ [1] for correct interpretation of 'during the fourth second'
 - ✓ [1] for obtaining $v_{\text{avg}} = \int_3^4 v(t) dt$
 - ✓ [1] for finding the exact average velocity
- During the fourth second means between $t = 3$ to $t = 4$.

$$\begin{aligned}
 \therefore v_{\text{avg}} &= \frac{x(4) - x(3)}{4 - 3} \\
 &= x(4) - x(3) \\
 &= \int_3^4 v(t) dt \\
 &= \int_3^4 \frac{(2t+1)^{\frac{3}{2}}}{3} - \frac{1}{3} dt \\
 &= \frac{1}{3} \int_3^4 (2t+1)^{\frac{3}{2}} - 1 dt \\
 &= \frac{1}{3} \left[\frac{(2t+1)^{\frac{5}{2}}}{\frac{5}{2} \times 2} - t \right]_3^4 \\
 &= \frac{1}{3} \left[\frac{(2t+1)^{\frac{5}{2}}}{5} - t \right]_3^4 \\
 &= \frac{1}{3} \left(\frac{223}{5} - \left(\frac{7^{\frac{5}{2}}}{5} - 3 \right) \right) \\
 &= \frac{1}{3} \left(\frac{238}{5} - \frac{7^{\frac{5}{2}}}{5} \right) \text{ m/s}
 \end{aligned}$$

Question 18 (Ho) (3 marks)

- ✓ [1] for correct overall shape of $f'(x)$
- ✓ [1] for correct location of $f'(x)$ at either O , $x = b$, OR $x = c$
- ✓ [1] for correct locations of $f'(x)$ at O , $x = a$, $x = b$, AND $x = c$



Question 19 (Lee)

(a) (3 marks)

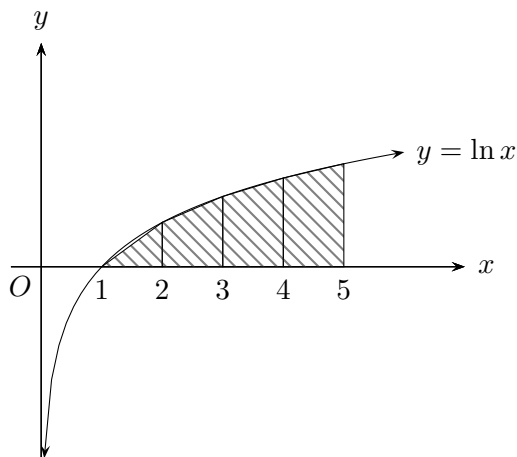
- ✓ [1] for correct x -values
- ✓ [1] for correct y -values
- ✓ [1] for final answer (rounding errors are ignored)

x	1	2	3	4	5
y	0	$\ln 2$	$\ln 3$	$\ln 4$	$\ln 5$

$$\begin{aligned}
 \therefore \int_1^5 \ln x dx &\approx \frac{1}{2} [\ln 5 + 2(\ln 2 + \ln 3 + \ln 4)] \\
 &= 3.98277... \\
 &= 3.9828 \text{ (4 d.p.)}
 \end{aligned}$$

(b) (2 marks)

- ✓ [1] for sketching $y = \ln x$ and trapezia between $x = 1$ to $x = 5$
- ✓ [1] for correct explanation



As the trapezia lie entirely beneath the graph of $y = \ln x$, the approximation found in part (a) is an under-approximation of $\int_1^5 \ln x dx$.

OR

As the trapezia do not completely cover the area represented by $\int_1^5 \ln x dx$, the approximation found in part (a) is an under-approximation.

OR

Any valid explanation.

Question 20 (Lee)

(a) (1 mark)

✓ [1] for correct expression of A_2

$$A_1 = M(1.0025)$$

$$\begin{aligned} A_2 &= [M(1.0025) + M](1.0025) \\ &= M(1.0025)^2 + M(1.0025) \end{aligned}$$

(b) (2 marks)

✓ [1] for obtaining correct expression of A_3

✓ [1] for correct proof

$$\begin{aligned} A_3 &= [M(1.0025)^2 + M(1.0025) + M](1.0025) \\ &= M(1.0025)^3 + M(1.0025)^2 + M(1.0025) \end{aligned}$$

$$\begin{aligned} \therefore A_n &= M(1.0025) + M(1.0025)^2 + M(1.0025)^3 + \cdots + M(1.0025)^n \\ &= M \left(\frac{1.0025(1.0025^n - 1)}{1.0025 - 1} \right) \\ &= 401M(1.0025^n - 1) \end{aligned}$$

(c) (2 marks)

✓ [1] for substituting $n = 24$ and $A_{24} = 15000$ ✓ [1] for correct value of M (no mark if M is rounded incorrectly)When $n = 24$, $A_n = \$15000$

$$\begin{aligned} \therefore 15000 &= 401M(1.0025^{24} - 1) \\ \therefore M &= 605.703... \end{aligned}$$

Therefore, he must invest \$606 per month.

Question 21 (Lee) (5 marks)

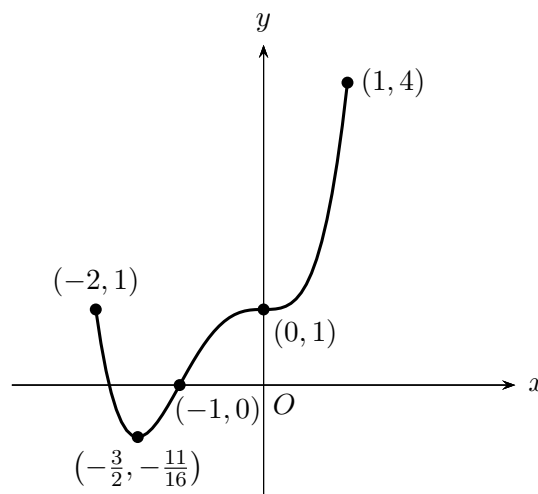
- ✓ [1] for determining the x values of the stationary points
- ✓ [1] for determining the nature of ALL $\therefore (-1, 0)$ is a point of inflection. stationary points
- ✓ [1] for determining the points of inflection
- ✓ [1] for correct sketch within the given domain, with ALL features labelled
- ✓ [1] for correct global maximum value

x	-2	-1	-0.5
$f''(x)$	24	0	-3
	+		-

$$\frac{dy}{dx} = 4x^3 + 6x^2$$

Let $\frac{dy}{dx} = 0$

$$\begin{aligned}\therefore 4x^3 + 6x^2 &= 0 \\ 2x^2(2x + 3) &= 0 \\ \therefore x &= 0, -\frac{3}{2}\end{aligned}$$



From the graph, global maximum = 4.

$$\frac{d^2y}{dx^2} = 12x^2 + 12x$$

When $x = 0$, $\frac{d^2y}{dx^2} = 0 \implies$ inconclusive; use table

x	-1	0	1
$f'(x)$	2	0	10
	$\nearrow$	$\rightarrow$	$\nearrow$

$\therefore (0, 1)$ is a horizontal point of inflexion.

When $x = -\frac{3}{2}$, $\frac{d^2y}{dx^2} = 9 > 0 \implies \left(-\frac{3}{2}, -\frac{11}{16}\right)$ is a minimum turning point.

Let $\frac{d^2y}{dx^2} = 0$

$$\begin{aligned}\therefore 12x^2 + 12x &= 0 \\ 12x(x + 1) &= 0 \\ \therefore x &= 0, -1\end{aligned}$$

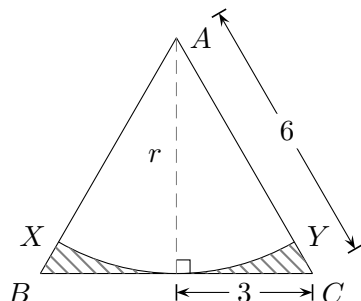
Already established there is a point of inflection at $x = 0$.

Check whether there is a point of inflection at $x = -1$:

Question 22 (Sun)

(a) (2 marks)

- ✓ [1] for substantial progress
- ✓ [1] for correct proof



$$\begin{aligned}
 r &= \sqrt{6^2 - 3^2} \\
 &= \sqrt{27} \\
 &= 3\sqrt{3} \text{ cm}
 \end{aligned}$$

(b) (1 mark)

- ✓ [1] for final answer

$$\begin{aligned}
 \text{arc } XY &= 3\sqrt{3} \times \frac{\pi}{3} \\
 &= \sqrt{3}\pi \text{ cm}
 \end{aligned}$$

(c) (3 marks)

- ✓ [1] for correct value of $A_{\triangle}$
- ✓ [1] for correct value of A_{sector}
- ✓ [1] for final answer

$$\begin{aligned}
 A_{\triangle} &= \frac{1}{2} \times 6 \times 3\sqrt{3} \\
 &= 9\sqrt{3} \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 A_{\text{sector}} &= \frac{1}{2} \times (3\sqrt{3})^2 \times \frac{\pi}{3} \\
 &= \frac{1}{2} \times 27 \times \frac{\pi}{3} \\
 &= \frac{9\pi}{2} \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \therefore A_{\text{shaded}} &= A_{\triangle} - A_{\text{sector}} \\
 &= 9\sqrt{3} - \frac{9\pi}{2} \text{ cm}^2
 \end{aligned}$$

Question 23 (Sun) (2 marks)

- ✓ [1] for using the identity $\sec^2 x = \tan^2 x + 1$
- ✓ [1] for correct values of $\tan x$

$$\begin{aligned}
 \tan^2 x + \sec^2 x &= 9 \\
 \tan^2 x + \tan^2 x + 1 &= 9 \\
 2\tan^2 x &= 8 \\
 \tan^2 x &= 4 \\
 \therefore \tan x &= \pm 2
 \end{aligned}$$

Question 24 (Sun) (2 marks)

- ✓ [1] for substantial progress
- ✓ [1] for correct proof

$$\text{LHS} = \frac{1}{\cos^2 \theta} + \frac{1}{\sin^2 \theta}$$

$$= \frac{\sin^2 \theta + \cos^2 \theta}{\cos^2 \theta \sin^2 \theta}$$

$$= \frac{1}{\cos^2 \theta \sin^2 \theta}$$

$$= \sec^2 \theta \operatorname{cosec}^2 \theta$$

$$= \text{RHS}$$

Question 25 (Sun)

(a) (2 marks)

- ✓ [1] for any two correct values of $P(X = x)$
- ✓ [1] for all four correct values of $P(X = x)$

x	-3	-2	1	3
$P(X = x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

(b) (2 marks)

✓ [1] for calculating $E(X)$ correctly

✓ [1] for final answer

$$\begin{aligned} E(X) &= -3 \times \frac{1}{8} - 2 \times \frac{3}{8} + 1 \times \frac{3}{8} + 3 \times \frac{1}{8} \\ &= -\frac{3}{8} \end{aligned}$$

$$\begin{aligned} \therefore \text{Expected loss after 1000 games} &= 1000 \times \frac{3}{8} \\ &= \$375 \end{aligned}$$

(c) (2 marks)

✓ [1] for simplification to $\frac{P(X > -2)}{P(X \geq -2)}$

✓ [1] for final answer

$$\begin{aligned} P(X > -2 | X \geq -2) &= \frac{P(X > -2 \cap X \geq -2)}{P(X \geq -2)} \\ &= \frac{P(X > -2)}{P(X \geq -2)} \\ &= \frac{\frac{4}{8}}{\frac{8}{8}} \\ &= \frac{4}{8} \end{aligned}$$

Question 26 (Lam)

(a) (2 marks)

- ✓ [1] for writing $\int_1^{14} \frac{k}{2t-1} dt = 1$
- ✓ [1] for correct proof

$$\begin{aligned}\int_1^{14} \frac{k}{2t-1} dt &= 1 \\ \frac{1}{2}k \int_1^{14} \frac{2}{2t-1} dt &= 1 \\ \frac{1}{2}k \left[\ln(2t-1) \right]_1^{14} &= 1 \\ \frac{1}{2}k (\ln 27 - \ln 1) &= 1 \\ \frac{1}{2}k \ln 27 &= 1 \\ \therefore k &= \frac{2}{\ln 27}\end{aligned}$$

(b) (2 marks)

- ✓ [1] for obtaining $\int_1^t \frac{k}{2t-1} dt = \frac{3}{4}$
- ✓ [1] for final answer

$$\begin{aligned}\int_1^t \frac{k}{2t-1} dt &= \frac{3}{4} \\ \frac{1}{2}k \left[\ln(2t-1) \right]_1^t &= \frac{3}{4} \\ \frac{1}{\ln 27} \left[\ln(2t-1) \right]_1^t &= \frac{3}{4} \\ \ln(2t-1) &= \frac{3 \ln 27}{4} \\ 2t-1 &= e^{\frac{3 \ln 27}{4}} \\ \therefore t &= 6.422\dots\end{aligned}$$

$\therefore$ After approximately 6.4 days.

Question 27 (Lam) (3 marks)

- ✓ [1] for any correct transformation
- ✓ [1] for any 3 correct transformations (incorrect order of transformations are ignored)
- ✓ [1] for correct sequence of transformations (correct order of transformations is required)

$$y = \frac{3}{-2\left(x - \frac{1}{2}\right)} + 5$$

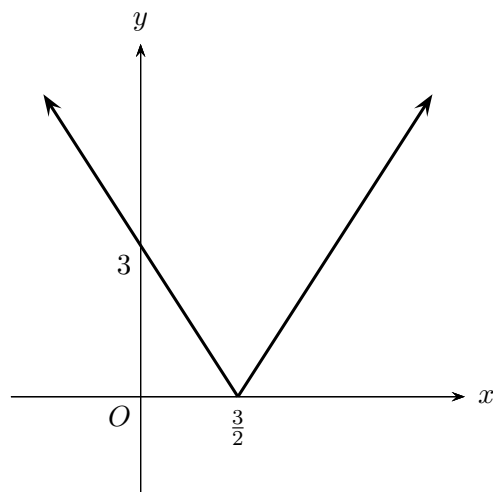
1. Reflect $y = \frac{1}{x}$ about the y -axis.
2. Dilate the resulting graph horizontally by a factor of $\frac{1}{2}$.
3. Translate the resulting graph horizontally to the right by $\frac{1}{2}$ units.
4. Dilate the resulting graph vertically by a factor of 3.
5. Translate the resulting graph vertically up by 5 units.

Other sequences of transformations are possible.

Question 28 (Lam)

(a) (2 marks)

- ✓ [1] for correct sketch
- ✓ [1] for correct axes intercepts



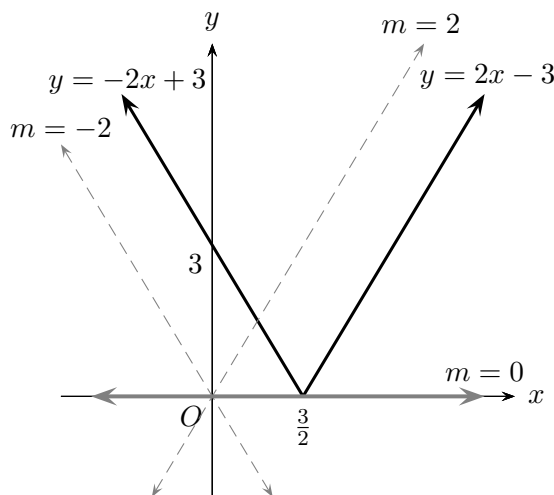
(b) (3 marks)

- ✓ [1] for substantial progress
- ✓ [1] for one correct value of x
- ✓ [1] for final answer

$$\begin{aligned}
 |6x - 9| &= 3 \\
 3|2x - 3| &= 3 \\
 |2x - 3| &= 1 \\
 2x - 3 &= \pm 1 \\
 2x &= 3 \pm 1 \\
 2x &= 4, 2 \\
 x &= 2, 1
 \end{aligned}$$

(c) (1 mark)

- ✓ [1] for final answer



From the graph, $m < -2$, $m \geq 2$, or $m = 0$.

Question 29 (Lam) (4 marks)

- ✓ [1] for obtaining $f(g(x)) = \frac{1}{\sin^2 x + \sin x - 2}$
- ✓ [1] for letting $\sin^2 x + \sin x - 2 = 0$
- ✓ [1] for obtaining $\sin x = -2, 1$
- ✓ [1] for final answer

$$\begin{aligned}
 y &= f(g(x)) \\
 &= f(\sin x) \\
 &= \frac{1}{\sin^2 x + \sin x - 2}
 \end{aligned}$$

 $\therefore$ Vertical asymptotes of $y = f(g(x))$ occur at

$$\begin{aligned}
 \sin^2 x + \sin x - 2 &= 0 \\
 (\sin x + 2)(\sin x - 1) &= 0 \\
 \therefore \sin x &= -2, 1 \\
 \therefore \sin x &= 1 \text{ (as } \sin x \neq -2) \\
 \therefore x &= \frac{\pi}{2}, \text{ for } x \in [0, 2\pi]
 \end{aligned}$$

Question 30 (Lam)

(a) (1 mark)

- ✓ [1] for final answer

$$P(\text{Sophie wins on her first draw}) = \frac{4}{7}.$$

(b) (2 marks)

- ✓ [1] for substantial progress
- ✓ [1] for final answer

$$\begin{aligned}
 &P(\text{Sophie wins in three or less of her turns}) \\
 &= P(\text{Sophie wins on her first draw}) \\
 &\quad + P(\text{Sophie wins on her second draw}) \\
 &\quad + P(\text{Sophie wins on her third draw}) \\
 &= \frac{4}{7} + \left(\frac{3}{7}\right)\left(\frac{4}{7}\right)\left(\frac{4}{7}\right) + \left(\frac{3}{7}\right)\left(\frac{4}{7}\right)\left(\frac{3}{7}\right)\left(\frac{4}{7}\right)\left(\frac{4}{7}\right) \\
 &= \frac{4}{7} + \left(\frac{3}{7}\right)\left(\frac{4}{7}\right)^2 + \left(\frac{3}{7}\right)^2\left(\frac{4}{7}\right)^3 \\
 &= \frac{4}{7} \left(1 + \frac{12}{49} + \left(\frac{12}{49}\right)^2\right) \\
 &= 0.7456...
 \end{aligned}$$

(c) (2 marks)

- ✓ [1] for establishing the correct infinite geometric series
- ✓ [1] for final answer

$$\begin{aligned}
 &P(\text{Sophie wins the game}) \\
 &= \frac{4}{7} \left(1 + \frac{12}{49} + \left(\frac{12}{49}\right)^2 + \dots\right) \\
 &= \frac{4}{7} \left(\frac{1}{1 - \frac{12}{49}}\right) \\
 &= \frac{28}{37}
 \end{aligned}$$

Question 31 (Lam) (3 marks)

- ✓ [1] for writing $A = \int_1^e (\ln(2x) - \ln x) dx$
- ✓ [1] for simplifying the integrand using log laws
- ✓ [1] for final answer

$$\begin{aligned}
 A &= \int_1^e (\ln(2x) - \ln x) dx \\
 &= \int_1^e \ln\left(\frac{2x}{x}\right) dx \\
 &= \int_1^e \ln 2 dx \\
 &= \left[x \ln 2 \right]_1^e \\
 &= e \ln 2 - \ln 2 \\
 &= (e - 1) \ln 2 \text{ units}^2
 \end{aligned}$$

Question 32 (Lam)

(a) (2 marks)

- ✓ [1] for obtaining correct expressions of $\sin \alpha$ AND d^2
- ✓ [1] for correct proof

$$\begin{aligned}
 I &= \frac{k \sin \alpha}{d^2} \\
 \sin \alpha &= \frac{y}{d} \\
 d^2 &= y^2 + a^2 \\
 \therefore I &= \frac{k \left(\frac{y}{d}\right)}{y^2 + a^2} \\
 &= \frac{ky}{d(y^2 + a^2)} \\
 &= \frac{ky}{\sqrt{y^2 + a^2} (y^2 + a^2)} \\
 &= \frac{ky}{(y^2 + a^2)^{\frac{3}{2}}}
 \end{aligned}$$

(b) (3 marks)

- ✓ [1] for finding $\frac{dI}{dx}$, or equivalent merit
- ✓ [1] for obtaining $y = 0, \pm \frac{a}{\sqrt{2}}$
- ✓ [1] for proving that $y = \frac{a}{\sqrt{2}}$ provides the maximum illumination to the circumference

$$I = \frac{k^2 y^2}{(y^2 + a^2)^3}$$

$$\begin{aligned}
 \frac{dI}{dy} &= \frac{(y^2 + a^2)^3 \cdot 2k^2 y - k^2 y^2 \cdot 3(y^2 + a^2)^2 \cdot 2y}{(y^2 + a^2)^6} \\
 &= \frac{2k^2 y (y^2 + a^2)^3 - 6k^2 y^3 (y^2 + a^2)^2}{(y^2 + a^2)^6} \\
 &= \frac{2k^2 y (y^2 + a^2)^2 [(y^2 + a^2) - 3y^2]}{(y^2 + a^2)^6} \\
 &= \frac{2k^2 y (a^2 - 2y^2)}{(y^2 + a^2)^4}
 \end{aligned}$$

Let $\frac{dI}{dy} = 0$

$$\therefore 2k^2 y (a^2 - 2y^2) = 0$$

$$\therefore y = 0, \pm \frac{a}{\sqrt{2}}$$

$$\therefore y = 0, \frac{a}{\sqrt{2}} \text{ (as } y \geq 0 \text{)}$$

But $y = 0$ provides no illumination to the circumference.

$$\therefore y = \frac{a}{\sqrt{2}}$$

y	$\frac{a}{2}$	$\frac{a}{\sqrt{2}}$	a
$\frac{dI}{dy}$	$\frac{2k^2 \left(\frac{a}{2}\right) \left(\frac{a^2}{2}\right)}{\left(\frac{a^2}{4} + a^2\right)^4}$	0	$\frac{2k^2 a (-a^2)}{(a^2 + a^2)^4}$

$\nearrow \qquad \longrightarrow \qquad \searrow$

$\therefore y = \frac{a}{\sqrt{2}}$ provides the maximum illumination to the circumference.